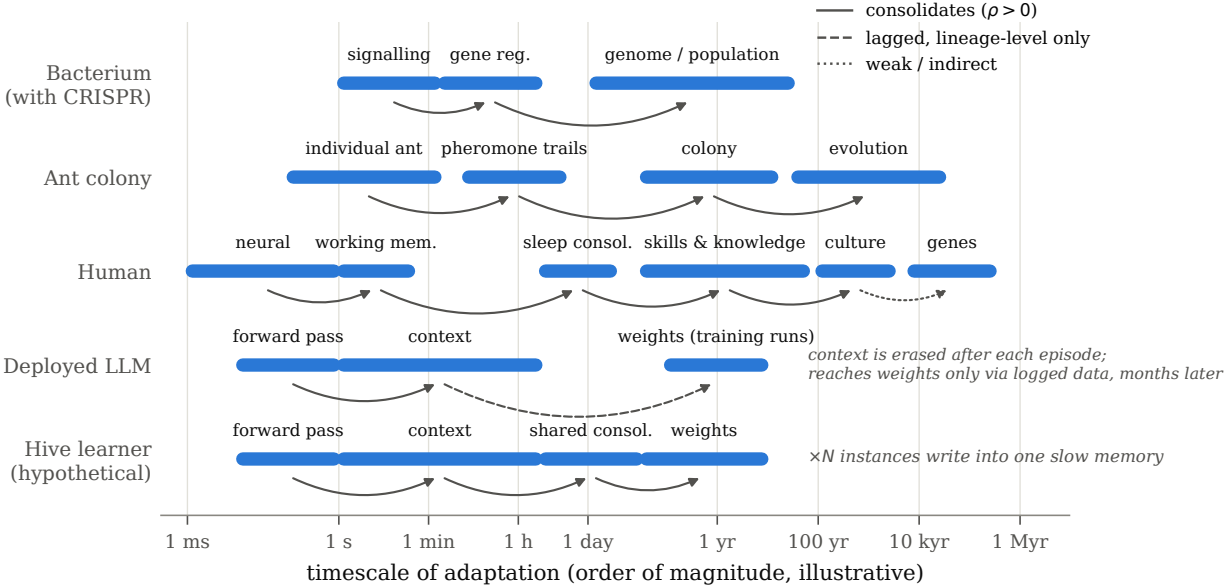
consolidates (ρ > 0)
lagged, lineage-level only
weak / indirect
Bacterium (with CRISPR)
signalling
gene reg.
genome / population
Ant colony
individual ant
pheromone trails
colony
evolution
Human
neural
working mem.
sleep consol.
skills & knowledge
culture
genes
Deployed LLM
forward pass
context
weights (training runs)
context is erased after each episode;
reaches weights only via logged data, months later
Hive learner (hypothetical)
forward pass
context
shared consol.
weights
×N instances write into one slow memory
1 ms
1 s
1 min
1 h
1 day
1 yr
100 yr
10 kyr
1 Myr
timescale of adaptation (order of magnitude, illustrative)